

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

SECTION 1. IDENTIFICATION

Product name : ELECTROPURE COPPER SULFATE SOL
Product code : 2200364-0055-4-000

Manufacturer or supplier's details

Company name of supplier : Atotech Deutschland GmbH

Address : Erasmusstrasse 20
Berlin 10553
Germany

Telephone : +4930349850

Company name of supplier : Atotech USA

Address : 1750 OVERVIEW DRIVE
ROCK HILL, SC, USA 29730

Telephone : +18038173500

Company name of supplier : Atotech Canada

Address : 1180 Corporate Drive
BURLINGTON L7L 5R6
Canada

Telephone : +19053320111

Prepared by
Product Safety Department (PSD): product-safety@atotech.com

Inquiries
E-mail address for a competent person responsible for the safety data sheet: product-safety@atotech.com

Emergency telephone number : CHEMTREC +18004249300

Transport Medical : Rocky Mountain Poison Control Center: 303-623-5716

Recommended use of the chemical and restrictions on use

Recommended use : Plating agents and metal surface treating agents
Surface treatment

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

Restrictions on use : For industrial use only.

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with 29 CFR 1910.1200

Acute toxicity (Oral)	: Category 4
Serious eye damage	: Category 1
Short-term (acute) aquatic hazard	: Category 1
Long-term (chronic) aquatic hazard	: Category 1

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H302 Harmful if swallowed.
H318 Causes serious eye damage.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P270 Do not eat, drink or smoke when using this product.
P273 Avoid release to the environment.
P280 Wear eye protection/ face protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER/ doctor.
P391 Collect spillage.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

None known.

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Chemical nature : Aqueous solution

Hazardous ingredients

Chemical name	CAS-No.	Concentration (% w/w)
Copper sulfate	7758-98-7	$\geq 10 - < 25$
Sulfuric acid	7664-93-9	$\geq 0.1 - < 1$

Actual concentration is withheld as a trade secret

This product may contain component(s) that are not listed under disclosure. All components not listed, do not contain hazardous materials above de minimus disclosure limits as defined by OSHA, NIOSH, ACGIH or Canadian WHMIS 2015 regulations and or guidelines. Please refer to other sections of the SDS for information on safety, health and environmental guidelines and precautions.

SECTION 4. FIRST AID MEASURES

General advice : Call a physician or poison control center immediately.
Show this safety data sheet to the doctor in attendance.

If inhaled : Move to fresh air.

In case of skin contact : In case of contact, immediately flush skin with plenty of water.
Take off contaminated clothing and shoes immediately.
Wash contaminated clothing before re-use.

In case of eye contact : In case of contact, immediately flush eyes with plenty of water
for at least 30 minutes.
Consult a physician.

If swallowed : If swallowed, call a poison control centre or doctor immediately.
Never give anything by mouth to an unconscious person.
Do not induce vomiting without medical advice.

Most important symptoms and effects, both acute and delayed : Harmful if swallowed.
Causes serious eye damage.

Protection of first aid responders : First Aid responders should pay attention to self-protection
and use the recommended protective clothing

Notes to physician : For specialist advice physicians should contact the Poison
Control Center.

ELECTROPURE COPPER SULFATE SOLVersion 5.0
SDS_US_GHSSDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

SECTION 5. FIRE-FIGHTING MEASURES

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : No information available.
- Hazardous combustion products : Sulfur oxides
Copper oxides
- Specific extinguishing methods : Use a water spray to cool fully closed containers.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
- Special protective equipment for fire-fighters : As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
Evacuate personnel to safe areas.
Keep people away from and upwind of spill/leak.
- Environmental precautions : Should not be released into the environment.
Local authorities should be advised if significant spillages cannot be contained.
- Methods and materials for containment and cleaning up : Avoid formation of aerosol.
Dam up.
Soak up with inert absorbent material.
Keep in suitable, closed containers for disposal.
Clean contaminated floors and objects thoroughly while observing environmental regulations.

SECTION 7. HANDLING AND STORAGE

- Advice on safe handling : For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Handle in accordance with good industrial hygiene and safety practice.
In case of insufficient ventilation, wear suitable respiratory equipment.
Avoid breathing mist or vapours.
- Conditions for safe storage : Keep containers tightly closed in a dry, cool and well-

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

ventilated place.

Do not freeze.

Recommended storage temperature : 7 - 40 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Copper sulfate	7758-98-7	TWA	1 mg/m ³ (Copper)	NIOSH REL
Sulfuric acid	7664-93-9	TWA (Thoracic particulate matter)	0.2 mg/m ³	ACGIH
		TWA	1 mg/m ³	NIOSH REL
		TWA	1 mg/m ³	OSHA Z-1
		TWA	1 mg/m ³	OSHA P0

Engineering measures : Ensure adequate ventilation, especially in confined areas.

Personal protective equipment

Respiratory protection : Use NIOSH approved respiratory protection.
In case of insufficient ventilation, wear suitable respiratory equipment.
When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

No personal respiratory protective equipment normally required.

Hand protection

Remarks : Impervious gloves

Eye protection : Tightly fitting safety goggles
Face-shield
Ensure that eyewash stations and safety showers are close to the workstation location.

Skin and body protection : Impervious clothing
Boots

Hygiene measures : Avoid contact with skin, eyes and clothing.
Wash hands before breaks and immediately after handling the product.
When using do not eat, drink or smoke.

ELECTROPURE COPPER SULFATE SOLVersion 5.0
SDS_US_GHSSDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: liquid
Color	: blue
Odor	: odorless
Odor Threshold	: Not relevant
pH	: < 2.0 (20 °C)
Melting point/freezing point	: not determined, mixture of various components
Initial boiling point and boiling range	: > 100 °C
Flash point	: does not flash
Evaporation rate	: not determined, mixture of various components
Flammability (solid, gas)	: not applicable for liquids
Upper explosion limit	: not determined, mixture of various components
Lower explosion limit	: not determined, mixture of various components
Vapor pressure	: ca. 23 hPa (20 °C)
Relative vapor density	: not determined, mixture of various components
Density	: 1.13 - 1.23 g/cm ³ (20 °C)
Solubility(ies) Water solubility	: completely miscible
Partition coefficient: n-octanol/water	: not determined, mixture of various components
Autoignition temperature	: not determined, stable under normal process conditions
Decomposition temperature	: not determined, stable under normal process conditions
Viscosity Viscosity, dynamic	: similar to water
Viscosity, kinematic	: similar to water
Oxidizing properties	: The substance or mixture is not classified as oxidizing.

SECTION 10. STABILITY AND REACTIVITY

ELECTROPURE COPPER SULFATE SOLVersion 5.0
SDS_US_GHSSDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

Reactivity	: None under normal processing.
Chemical stability	: Stable under recommended storage conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: To avoid thermal decomposition, do not overheat.
Incompatible materials	: Bases
Hazardous decomposition products	: No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION**Information on likely routes of exposure**

Eye contact

Acute toxicity

Harmful if swallowed.

Product:

Acute oral toxicity	: Acute toxicity estimate: 1,950 mg/kg Method: Calculation method
Acute dermal toxicity	: Acute toxicity estimate: > 5,000 mg/kg Method: Calculation method

Remark: The acute toxicity estimate (ATE) of the ingredients are derived using the LD50/LC50 values where available.

Components:**Copper sulfate:**

Acute oral toxicity	: LD50 Oral (Rat): 308 mg/kg Method: OECD Test Guideline 401 Remarks: The value is calculated ECHA registration dossier
Acute dermal toxicity	: LD50 Dermal (Rabbit): 2,002 mg/kg Remarks: published study from scientific journal LD50 Dermal (Rat): > 2,000 mg/kg Method: OECD Test Guideline 402 Remarks: information from raw material supplier see user defined free text

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Serious eye damage/eye irritation

Causes serious eye damage.

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

Product:

Remarks: May cause irreversible eye damage such as corneal damage and blindness.

Components:

Copper sulfate:

Species: Rabbit

Result: Irreversible effects on the eye

Assessment: Risk of serious damage to eyes.

Method: OECD Test Guideline 405

Remarks: ECHA registration dossier

Respiratory or skin sensitization

Skin sensitisation

Based on available data, the classification criteria are not met.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Components:

Copper sulfate:

Species: Guinea pig

Method: OECD Test Guideline 406

Result: negative

Remarks: information from raw material supplier

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Carcinogenicity

Based on available data, the classification criteria are not met.

Product:

Remarks: The IARC has classified "Strong inorganic acid mists containing sulfuric acid" as a Category 1 carcinogen, a substance that is "carcinogenic to humans". This classification is for strong inorganic acid mists only and does not apply to sulfuric acid or sulfuric acid solutions.

IARC

Group 1: Carcinogenic to humans

Sulfuric acid

7664-93-9

OSHA specified

No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.

NTP

Known to be human carcinogen

Sulfuric acid

7664-93-9

ELECTROPURE COPPER SULFATE SOLVersion 5.0
SDS_US_GHSSDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

Reproductive toxicity

Based on available data, the classification criteria are not met.

STOT - single exposure

Based on available data, the classification criteria are not met.

STOT - repeated exposure

Based on available data, the classification criteria are not met.

Aspiration toxicity

Based on available data, the classification criteria are not met.

Further information**Product:**

Remarks: No data available

SECTION 12. ECOLOGICAL INFORMATION**Ecotoxicity****Components:****Copper sulfate:**

Toxicity to fish : LC50 (Fish): 0.11 mg/l
Exposure time: 96 h
Remarks: information from raw material supplier

Toxicity to daphnia and : EC50 (Daphnia magna (Water flea)): 0.02 mg/l
other aquatic invertebrates
Exposure time: 48 h
Remarks: information from raw material supplier

Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects**Product:**

No data available

SECTION 13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : Dispose of in accordance with local regulations.
Dispose of wastes in an approved waste disposal facility.

Contaminated packaging : Empty containers should be taken to an approved waste han-

ELECTROPURE COPPER SULFATE SOLVersion 5.0
SDS_US_GHSSDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

dling site for recycling or disposal.

SECTION 14. TRANSPORT INFORMATION**International Regulations****IATA-DGR**

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
Technical name(s) : (Copper sulfate)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
Technical name(s) : N.O.S.
(Copper sulfate)
Class : 9
Packing group : III
Labels : 9
EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Domestic regulation**DOT / 49 CFR**

UN/ID/NA number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S.
(Copper sulfate)
Class : 9
Packing group : III
Labels : CLASS 9
ERG Code : 171
Marine pollutant : yes (Copper sulfate)
Remarks : Above applies only to containers over 119 gallons or 450 liters. Not regulated if shipped in packages less than or equal to 119 gallons (450 liters).

SECTION 15. REGULATORY INFORMATION

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

TSCA 5a : No substances are subject to a Significant New Use Rule.

TSCA 12b : No substances are subject to TSCA 12(b) export notification requirements.

DEA : Not applicable

EPCRA - Emergency Planning and Community Right-to-Know Act

CERCLA Reportable Quantity

Components	CAS-No.	Component RQ (lbs)	Calculated product RQ (lbs)
Copper sulfate	7758-98-7	10	63
Sulfuric acid	7664-93-9	1000	*

*: Calculated RQ exceeds reasonably attainable upper limit.

SARA 304 Extremely Hazardous Substances Reportable Quantity

Components	CAS-No.	Component RQ (lbs)	Calculated product RQ (lbs)
Sulfuric acid	7664-93-9	1000	*

*: Calculated RQ exceeds reasonably attainable upper limit.

SARA 311/312 Hazards : Acute toxicity (any route of exposure)
Serious eye damage or eye irritation

SARA 302 : No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

SARA 313 : The following components are subject to reporting levels established by SARA Title III, Section 313:

Copper sulfate 7758-98-7

US State Regulations

Massachusetts Right To Know

Copper sulfate	7758-98-7	10 - 25 %
Sulfuric acid	7664-93-9	0.1 - 1 %

Pennsylvania Right To Know

Copper sulfate	7758-98-7	10 - 25 %
----------------	-----------	-----------

New Jersey Right To Know

Copper sulfate	7758-98-7	10 - 25 %
Sulfuric acid	7664-93-9	0.1 - 1 %

California Prop. 65

WARNING: This product can expose you to chemicals including Sulfuric acid, which is/are known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

Remarks: Components which are only displayed in Section 15 are being reported for local regulatory purposes. These components are not displayed in Section 3 due to one or more of the following conditions being met: being present in the product at concentration(s) below threshold limit values for reporting, not considered hazardous materials, health hazards or because they do not contribute to the overall GHS Classification of the final product as required by OSHA HazCom 2012 final rule (29 CFR 1910.1200).

Substances currently restricted by WEEE/RoHS (European Directives 2015/863/EU, 2012/19/EU and 2011/65/EU) or ELV (European Directive 2000/53/EC):

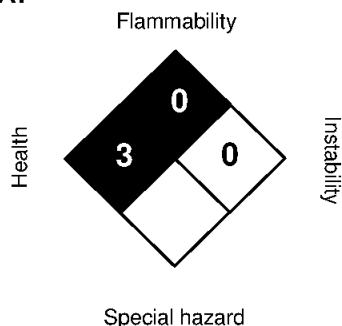
PBDE	PBB	CrVI	Hg	Pb	Cd
-	-	-	-	-	-
Phthalates:		DEHP	BBP	DBP	DIBP
		-	-	-	-

Please note: Current legislation according to WEEE/RoHS or ELV restricting the use of certain substances applies to "homogeneous material" in finished articles being supplied to the market. Substances deposited during surface finishing may have a composition (weight percent) higher than the weight percent of the substance in the operating solution from which the deposit is made. Atotech encourages its customers to implement systems to ensure their finished products comply with the regulations in force.

SECTION 16. OTHER INFORMATION

Further information

NFPA:



Full text of other abbreviations

ACGIH	:	USA. ACGIH Threshold Limit Values (TLV)
NIOSH REL	:	USA. NIOSH Recommended Exposure Limits
OSHA P0	:	USA. OSHA - TABLE Z-1 Limits for Air Contaminants - 1910.1000
OSHA Z-1	:	USA. Occupational Exposure Limits (OSHA) - Table Z-1 Lim-

ELECTROPURE COPPER SULFATE SOL

Version 5.0
SDS_US_GHS

SDS Number: 2200364-0055-4-
000

Revision Date: 05/22/2020

its for Air Contaminants
ACGIH / TWA : 8-hour, time-weighted average
NIOSH REL / TWA : Time-weighted average concentration for up to a 10-hour
workday during a 40-hour workweek
OSHA P0 / TWA : 8-hour time weighted average
OSHA Z-1 / TWA : 8-hour time weighted average
(Q)SAR - (Quantitative) Structure Activity Relationship; ASTM - American Society for the Testing
of Materials; bw - Body weight; DIN - Standard of the German Institute for Standardisation; ECx -
Concentration associated with x% response; ELx - Loading rate associated with x% response;
EmS - Emergency Schedule; ErCx - Concentration associated with x% growth rate response;
GHS - Globally Harmonized System; IARC - International Agency for Research on Cancer; IATA -
International Air Transport Association; IBC - International Code for the Construction and Equip-
ment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentra-
tion; ICAO - International Civil Aviation Organization; IMDG - International Maritime Dangerous
Goods; IMO - International Maritime Organization; ISO - International Organisation for Standardi-
zation; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a
test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of
Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect
Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect
Loading Rate; OECD - Organization for Economic Co-operation and Development; OPPTS - Of-
fice of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic
substance; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the
Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals;
SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; UN - United
Nations; vPvB - Very Persistent and Very Bioaccumulative; CERCLA - Comprehensive Environ-
mental Response, Compensation, and Liability Act; DOT - Department of Transportation; EHS -
Extremely Hazardous Substance; HMIS - Hazardous Materials Identification System; MSHA -
Mine Safety and Health Administration; NFPA - National Fire Protection Association; RCRA - Re-
source Conservation and Recovery Act; RQ - Reportable Quantity; SARA - Superfund Amend-
ments and Reauthorization Act; CMR - Carcinogen, Mutagen or Reproductive Toxicant; GLP -
Good Laboratory Practice; ERG - Emergency Response Guide; NTP - National Toxicology Pro-
gram; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods

Revision Date : 05/22/2020

The information provided in this Safety Data Sheet is correct to the best of our knowledge, infor-
mation and belief at the date of its publication. The information given is designed only as a guid-
ance for safe handling, use, processing, storage, transportation, disposal and release and is not
to be considered a warranty or quality specification. The information relates only to the specific
material designated and may not be valid for such material used in combination with any other
materials or in any process, unless specified in the text.

US / EN